→ mostly followed by Rheumatologist

Connective tissue disorders

primary care block/dermatology

Dr.fenik j. Yawar
Board certified dermatologist

Content

- Introduction
- Lupus erythematosus
- dermatomyositis
- Scleroderma
- Morphe~~d~~a

Learning outcomes

- To know different connective tissue disorders and their skin manifestation
- To be able to diagnose them and differentiate them
- To be able to manage these disorders

Introduction

- The main feature of these conditions is inflammation in the connective tissue which leads to dermal atrophy or sclerosis, arthritis, and sometimes to abnormalities in other organs.
- In addition, antibodies form against normal tissues and cellular components; therefore they are classified as autoimmune diseases.

systemic sclerosis

- Lupus erythematosus, dermatomyositis, scleroderma, rheumatoid artheritis, Sjögren syndrome, eosinophilic fasciitis and relapsing polychondritis.

Lupus erythematosus → always Photosensitive

This disorder has a spectrum, ranging from purely cutaneous type (discoid LE), through patterns associated with some internal problems (disseminated discoid LE and subacute cutaneous LE), to severe multisystem disease (systemic lupus erythematosus SLE).

Systemic lupus erythematosus

Etiology

- Hereditary factors, such as complement deficiency and certain HLA types; increase susceptibility.
- Particles looking like viruses have been seen in endothelial cells and in other tissues.
- Autoantibodies to DNA, nuclear proteins and to other normal antigens, which points to autoimmune cause.
- Exposure to sunlight and artificial ultraviolet radiation (UVR), pregnancy and infection may
  not causing it } → causation need genetic susceptibility
  precipitate the disease or leads to flare-ups.
- larger dose more precipitation
  Some drugs such as hydralazine and procainamide trigger SLE in a dose dependant manner, whereas oral contraceptives, anticonvulsants, minocycline and captopril precipitate the disease occasionally.
  in dose - independent

* precipitate means worsen or unmask

Clinical presentation

The onset is acute.

-Affects females more than males (8:1).

-The classic rash is erythema of cheeks and nose in shape of butterfly, with facial swelling.

-Occasionally few blisters may be seen.

→ Scarring Alopecia in scalp

-Some patients develop widespread discoid papulosquamous plaques very like those of discoid LE; about 20% of patients have no skin disease at any stage.

-Other dermatological features include periungual telangiectasia, erythema over the

non scarring → due to telogenic effluvium

digits, hair fall specially in the frontal area of the scalp and photosensitivity.

-ulcers may occur on the palate, tongue or buccal mucosa.

may be painless

malar rash

Raynaud phenomenon

Course:

-SLE may be associated with fever, arthritis, nephritis, polyarteritis, pleurisy, pneumonia, pericarditis, myocarditis and involvement of CNS.

-Renal involvement suggests poorer prognosis.

↳ hematuria, proteinuria, cast, proliferative glomerulonephritis

Complications:

The skin disease may cause scarring or hypopigmentation, but the main danger lie with damage to other organs and side effects of drugs, especially systemic steroid.

Criteria for diagnosis

→ one being ANA +

Must have four out of them:

1. Malar rash
2. Discoid plaques.
3. Photosensitivity.
4. non-recurrent Mouth ulcer.
5. Arthritis.
6. Serositis.
7. Renal disorder.
8. Neurological disorder.
9. Hematological disorder.
10. Immunological disorder. → anti-DNA or anti-smith or anti-phospholipid
11. Antinuclear antibody. → ANA

Investigations

usually not needed

last resource

-Skin biopsy and immunofluorescence which show fibrillar or granular deposits of IgG, IgM, IgA and/or C3 alone in basement membrane zone.

-Hematology: anemia, decrease WBC count, thrombocytopenia and raised ESR., normal CRP → ↑ in infection / serositis

-Urine analysis: protienura, hematuria and cast.

-Immunology:

*ANA: +ve in 100% of cases.

*Double stranded DNA: +ve in 50-70% of cases.

*Anti Ro and anti La in 20% of cases.

indicate co-association with anti-phospholipid syndrome

*Cardiolipin +ve in subsets with recurrent abortion, thrombosis, livido and skin necrosis.

*Anti Sm in 30% of cases.

-Tests for function of other organs: as indicated by history but always test kidney and liver functions.

Treatment

- -Bed rest during exacerbations.
- -Systemic steroid: large doses of prednisolone needed as aided to achieve control, as assessed by symptoms, signs, ESR, total complement levels and tests of other organs. The dose is then reduced to the smallest that suppress the disease.
- -Immunosupressive agents such as azathioprine and cyclophosphamide.
- ex: hydroxychloroquine
  -Antimalarial drugs may help some patients with rash + photosensitivity as may sunscreens.
- -Intermittent IV infusion of gamma globulin show promise.
- -Other drugs like antihypertensive therapy or anticonvulsants may be needed. according to organ involvement

Subacute cutaneous lupus erythematous

-Less severe than acute SLE.
-Its cause is unknown but may involve an antibody-dependant cellular cytotoxic attack on basal cells by K cells bridged by antibody to Ro (SS-A) antigen.
-Patients are often photosensitive.
look like psoriasis , polycyclin lesion
-Skin lesions are sharply demarkated scaling psoriasiform plaques, sometimes annular, lying on the forehead, nose, cheeks, chest, hands and extensor surfaces of the arms. -They are symmetrical and are hard to tell from discoid LE or SLE with widespread discoid lesions.

Course:
The course is prolonged, the skin lesions are slow to clear, but in contrast to discoid LE, do so with little or no scarring.

Complications:
-Systemic disease is frequent but not serious.
-Children born to mothers who have this condition are liable to neonatal LE, with annular skin lesions and permanent heart block.

Investigations:

They should be investigated as those of SLE.

Immunological tests:

*ANA +ve in up to 80% of cases.

*Anti Ro (SSA) and La (SSB) +ve in 60% of cases.

Treatment:

-They do better with antimalarials such as hydroxychloroquine than SLE.

-Oral retinoids are effective in some cases.

-Systemic steroid may be needed too.

*may cause hypo-pigmentation

Discoid lupus erythematous

-Most common form of LE, occurs in young adults with women outnumbering men by 2:1.
-Patients may have one or two plaques only, or many in several areas.
-The cause is also unknown, but UVR is one factor.

Presentation: → usually only skin involvement
-Lesionds begin as dull red macules or plaques, show erythema, adherent scaling, follicular plugging, scarring and atrophy, telangiectasis, hyopigmentation and a peripheral zone of hyperpigmentation.
-They are well demarcated and lie mostly on sun exposed skin of the scalp, ears and face.
-There are 2 types: localized DLE and generalized DLE.

Course:
-The disease may spread, but in about half of cases the disease goes to remission over the course of several years.
-Scarring is common and hair may lost permanently if there is scarring in the scalp.
-Whiteness remains after the inflammation has cleared, and hypopigmentation is common in dark skinned patients.
-Discoid LE rarely progress to SLE.

Investigations:

-Screening for SLE and internal disease is still worthwhile.
-Skin biopsy is most helpful if taken from an untreated plaque where appendages are still present.
-Direct immunofluorescence show deposits of IgG, IgM, IgA and C3 at the basement membrane zone.
-Blood tests are usually normal but 35% of patients have +ve ANA.

Treatment:

-Sun avoidance is important.
-Lesions need potent or very potent topical steroid even on the face as the risk of scarring is worse than that of atrophy, they should be applied twice daily until the lesions disappear, or side effects such as atrophy develop; weaker preparations are then used for maintenance.
-If no response then intralesional steroid injection can be used.
-Stubborn and widespread lesions often do well with antimalarials. or systemic steroids
-Oral retinoids and thalidomides can help stubborn cases.

Dermatomyositis

- It is a subset of polymyositis with distinctive skin changes.
- -There are juvenile and adult types.
- -The cause is unknown but an autoimmune mechanism seems likely. -Autoantibodies to striated muscles are found.
- -When starting after age of 40, dermatomyositis may signal an internal malignancy.
- -Serological evidence of toxoplasmosis was found in one series.

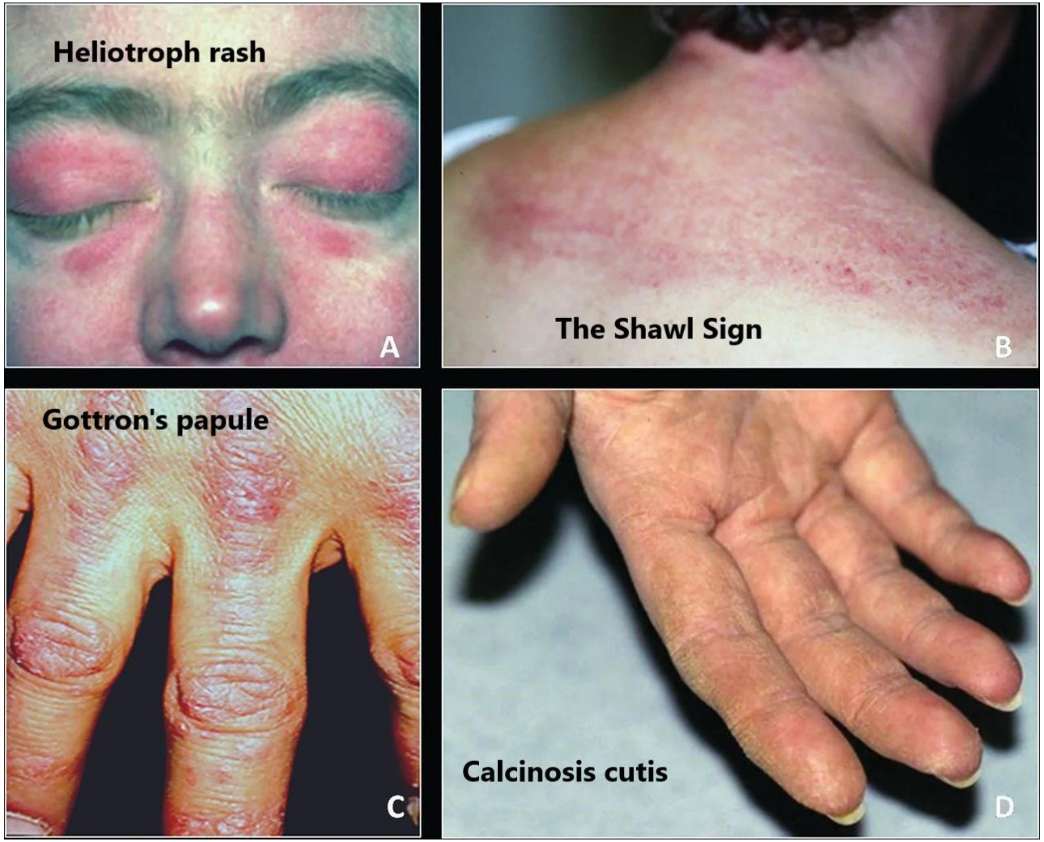
Clinical presentation

violet

- Typical patients have a faint lilac discoloration around their eyes (heliotrope).
- Malar erythema and edema and sometimes less striking erythema of the neck and presternal area.
- Most patients also develop lilac slightly atrophic papules over the knuckles of fingers (Gottron's papules), streaks of erythema over the extensor tendons of the hand, periaungual telangiectasia and ragged cuticles.

- The skin signs usually appear as the same time as the muscle symptoms but occasionally months or years earlier.
- Many, but not all, patients have weakness of proximal muscles. Climbing stairs, getting up from chairs and combing the hair become difficult.

*shawl sign



Shawl sign

hair rash

Calcinosis

Peri-ungual Telengatica

Course:

In children the disorder is self-limited, but in adults may be prolonged and progressive.
Raynaud's phenomenon, arthralgia, dysphagia and calcinosis may follow.

Complications:

Myositis may lead to permanent weakness and immobility, and inflammation to contractures or cutaneous calcinosis. Some die from progressive and severe myopathy.

Investigations:

Adult dermatomyositis requires a search for an underlying malignancy.
Muscle enzymes such as aldolase and creatinine phosphokinase (CPK) are elevated.
Electromyography (EMG) detect muscle abnormalities.
Muscle biopsy shows muscle inflammation and destruction.
ESR is often normal.
ANA may not be detected.
Toxoplasmosis should be excluded by serology.

Treatment

- High dose prednisolone (60 mg/day) is the cornerstone of treatment and protect the muscles from destruction
- Immunosuppressive agents such as azathioprine also help to control the condition and to reduce the high steroid dose.
- Cyclosporine and methotrexate are alternatives.
- Intravenous gamma globulin infusion seem promising.

systemic sclerosis

In this disorder the skin becomes hard as connective tissue thicken.
In the dermis the fibroblasts proliferate and produce more hyaluronic acid and type I collagen, there will be also intemal thickening of arterioles and arteries. These processes also involve gut, lungs, kidneys and heart leading to their dysfunction and death.

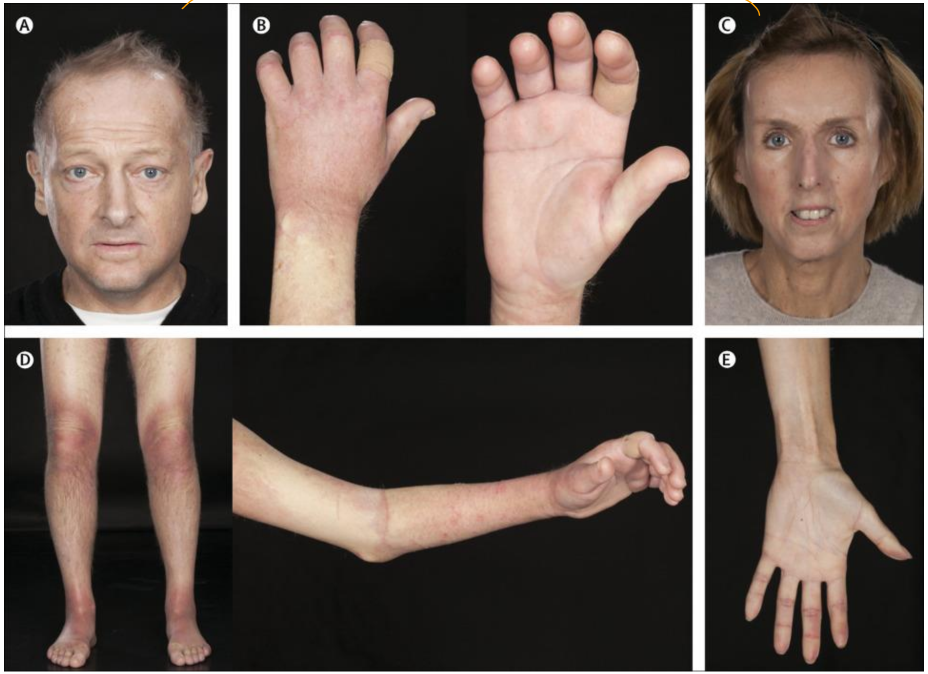
Clinical presentation

- Most patients suffer from raynaud's phenomenon and sclerodactyly. -Their fingers become immobile, hard, shiny and my ulcerate..
- Some become hyperpigmented and itch early in their disease, peri-ungual telangiectasis is common.
- Forehead creases accentuate (increase), the nose becomes pinched beak-like, small mouth with wrinkles radiate from it, and there will be diffuse hyperpigmentation and mat-like telangiectasia in the face.

- Abnormalities of the gut include; dysphagia, esophagitis, constipation, diarrhea, and malabsorption.
- Fibrosis of lungs leads to dyspnea and fibrosis of heart to congetive heart failure.
- Kidney involvement has grave prognosis.

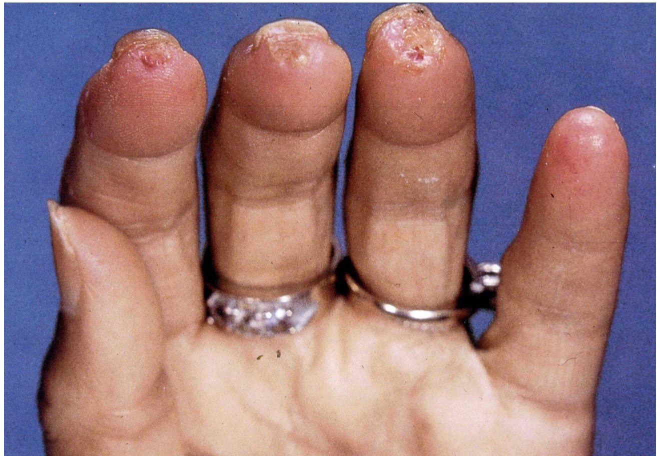
Typical face of SSc

Raynaud



Typical face

Raynaud + peri-ungual telangiectasia



Investigation:

-Diagnosis is made clinically.
-Biopsy of skin show sclerosis.
-ANA is +ve in 90% of cases which is speckled and nucleolar.
-Anti-centromere antibody is +ve in 50% of cases.
-Scleroderma associated antibody Scl-70 is worthwhile.
-Evaluation of kidney, heart, lung, joint and muscle.
-Avoid barium studies as obstruction may follow poor evacuation.

Treatment:

-Treatment is unsatisfactory. not curable
-Calcium channel blocker nifedipine may help Raynaud's phenomenon.
-Systemic steroids, antimalarials and long term penicillin are used but are not of proven value.
-D-penicillamine is used with many side effects.

Morphea

-This is a localized form of scleroderma with pale indurated plaques on the skin but no internal sclerosis.

-Many plaques are surrounded by violaceous halo

-Its prognosis is good, and the fibrosis slowly clears leaving slight depression and hypopigmentation. may leave scar behind

-A rare type may lead to arrest of growth of the underlying bone leading to facial asymmetry or short limb.

-Treatment works slowly and include topical steroid, calcipotriol, NSAID, UVA, PUVA and hydroxychloroquine.

facial assymetry

morphea